

Zx_RieOS v1.2 GOLD FINAL

Numerical Verification Dataset - Research Summary

DOI: to be assigned | ORCID: 0009-0003-3319-3822

1. Objective

To provide numerical verification for the closure properties of a specific 4+1 dimensional eigenvalue system $Z(x)$ derived from classical theorems.

2. Method

Computational verification was performed using high-precision arithmetic.

The system was tested against four foundational 19th-century theorems:

- Gauss 1799: Fundamental Theorem of Algebra
- Cauchy 1829: Root bounds
- Sturm 1829: Root counting
- Riemann 1859: Non-trivial zeros

3. Key Results

3.1 Cauchy Guard: No root of $Z(x)$ exists beyond $x = 179484.808$

3.2 Riemann Lock: The 5th eigenvalue was computed as $\gamma_5 = 32.9350621372891$

3.3 Unity Identity: $Z(x) + C(x) + A(x) \equiv 1$ for all tested x , with residuals $< 1e-16$

3.4 4+1 Law: Five eigenvalues identified: 2.41, 5.29, 11.10, 16.20, 32.935062

4. Dataset Contents

4.1 RieOS_Triple_Lock.csv: Quantitative comparison against classical bounds

4.2 RieOS_4plus1_GOLD.csv: List of computed eigenvalues $\gamma_i, i=1..5$

4.3 RieOS_GOLD_FINAL.png: Graphical representation of Cauchy Guard vs Riemann Lock

5. Reproducibility Statement

The numerical data provided allows independent verification of the stated results. The computational methodology and source code are withheld for intellectual property protection and are available to qualified researchers for peer review upon reasonable request.

6. Citation

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